॥ वसुधैव कुटुम्बकम् ॥

Symbiosis

Institue of technology

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ASSIGNMENT No: 3

TITLE: Develop a Java application demonstrating method overloading and the use of static variables, methods in class design.

Problem Statement: Develop a Java application demonstrating method overloading and the use of static variables, methods in class design.

Exercise 3.1 - Develop a Calculator program using overloaded methods for addition of integers and decimals. Use a static variable to count calculations.

Program –

```
class Calculator {
```

```java
// Static variable
static int totalCalculations = 0;

// Constructor
Calculator() {
   totalCalculations++;
}

// Overloaded methods
int add(int x, int y) {
   return x + y;
}

double add(double x, double y) {
   return x + y;
}

int add(int x, int y, int z) {
   return x + y + z;
}

// Static method
static void showCount() {
   System.out.println("Total Calculator Objects: " + totalCalculations);
}

public static void main(String[] args) {

   Calculator calc1 = new Calculator();
   Calculator calc2 = new Calculator();

   System.out.println("Sum of two integers : " + calc1.add(15, 25));
```

```
    System.out.println("Sum of two decimals : " + calc1.add(12.5, 18.5));
    System.out.println("Sum of three integers : " + calc1.add(10, 20, 30));

    showCount();
  }
}
```

Output :

```
PS C:\Users\HP> cd Documents\java
PS C:\Users\HP\Documents\java> javac Calculator.java
PS C:\Users\HP\Documents\java> java  Calculator
Sum of two integers : 40
Sum of two decimals : 31.0
Sum of three integers : 60
Total Calculator Objects: 2
PS C:\Users\HP\Documents\java>
```

EXERCISE 3.2 : Develop a Restaurant Billing Application where overloaded methods calculate bills for dine-in, takeaway, and delivery orders, while static variables track total orders.

Program –

```
class Calculator {

  // Static variable
  static int totalCalculations = 0;

  // Constructor
  Calculator() {
    totalCalculations++;
  }
```

```java
    // Overloaded methods
    int add(int x, int y) {
        return x + y;
    }

    double add(double x, double y) {
        return x + y;
    }

    int add(int x, int y, int z) {
        return x + y + z;
    }

    // Static method
    static void showCount() {
        System.out.println("Total Calculator Objects: " + totalCalculations);
    }

    public static void main(String[] args) {

        Calculator calc1 = new Calculator();
        Calculator calc2 = new Calculator();

        System.out.println("Sum of two integers : " + calc1.add(15, 25));
        System.out.println("Sum of two decimals : " + calc1.add(12.5, 18.5));
        System.out.println("Sum of three integers : " + calc1.add(10, 20, 30));

        showCount();
    }
}
```

Output :

```
PS C:\Users\HP\Documents\java> javac Restaurant.java
PS C:\Users\HP\Documents\java> java Restaurant
Dine-in Bill       : 450.0
Takeaway Bill      : 475.0
Delivery Bill      : 515.0
Total Orders Processed: 3
PS C:\Users\HP\Documents\java>
```

Conclusion : This assignment demonstrated the implementation of method overloading and the use of static variables and static methods in Java. Method overloading allowed the same method to perform different operations based on different parameter lists, while static members were used to share common data among all objects. The assignment improved the understanding of code reusability, flexibility, and efficient class design in object-oriented programming